Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Job Had Four Bros
Project / Solution Name	CoSolve
Name (Leader)	Wong Ye Yang
Email & Phone (Leader)	yeyangwong@gmail.com | 0103714818
University Affiliation(s)	Sunway College
Live Prototype URL	https://co-solve-complaint-resolution.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/JingHao0529/CoSolve

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

CoSolve is a web-based complaint-resolution workspace. A React and TypeScript frontend lets staff receive and review complaints. A Node.js and Express API stores case data through Drizzle ORM, records audit events, and coordinates AI services. The API sends complaint text to a Runpod Serverless endpoint hosting a fine-tuned DistilBERT sentiment model. The endpoint returns sentiment, confidence, a human-review flag, and a model version. A Grafilab OpenAI-compatible language-model API is used for structured assistance such as response drafting. Staff remain responsible for reviewing classifications, recommendations, compensation decisions, and outgoing messages. Authentication is provided through Clerk. The prototype is developed and hosted through Replit, with service credentials held in environment secrets rather than exposed in source code.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript	Frontend and backend application language	Apache-2.0
React	Web user interface	MIT
Node.js and Express	API server and routing	MIT
Drizzle ORM	Database access and schema management	Apache-2.0
Hugging Face Transformers, Datasets, Accelerate	Model fine-tuning and inference	Apache-2.0
PyTorch	Model-training runtime	BSD-style licence

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Runpod Serverless API	Hosted inference for the fine-tuned DistilBERT sentiment model	Paid usage / project account
Grafilab OpenAI-compatible API	Language-model assistance for support workflows and response drafts	Account-based usage
Clerk API	Staff authentication	Development tier / to be confirmed

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Web development environment and prototype hosting	Project workspace; region to be confirmed
Runpod	GPU training and serverless model inference	Project account; data centre selected through Runpod
Clerk	Authentication service	Development configuration

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DistilBERT sentiment-distilbert-v1	Fine-tuned by team using Hugging Face / Runpod	Fine-tuned	Base model: Apache-2.0
Grafilab-hosted language model	Grafilab	Pre-trained API model	Provider terms; exact model ID to be confirmed

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Womens Clothing E-Commerce Reviews	https://www.kaggle.com/datasets/nicapotato/womens-ecommerce-clothing-reviews	Public	Kaggle dataset listing; licence to be confirmed	N
Malaysia Restaurant Review Datasets	https://www.kaggle.com/datasets/choonkhong/malaysia-restaurant-review-datasets	Public	Kaggle dataset listing; licence to be confirmed	N / to be verified

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Womens Clothing E-Commerce Reviews	https://www.kaggle.com/datasets/nicapotato/womens-ecommerce-clothing-reviews	Public	Kaggle dataset listing; licence to be confirmed	N
Malaysia Restaurant Review Datasets	https://www.kaggle.com/datasets/choonkhong/malaysia-restaurant-review-datasets	Public	Kaggle dataset listing; licence to be confirmed	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT / Codex	Code assistance, debugging, deployment guidance and documentation	Moderate
Replit Agent	Code assistance and project iteration	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

ChatGPT / Codex and Replit Agent were used as development assistants. The team reviewed the generated suggestions, selected the final architecture, configured services and secrets, trained and evaluated the sentiment model, tested workflows, and remains responsible for the submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Development platform and prototype hosting	Used as the project workspace to develop, run, and demonstrate the full-stack CoSolve prototype, including frontend, backend, environment secrets, and iterative testing.
Runpod	GPU compute and serverless inference	Used to fine-tune the DistilBERT sentiment model on review datasets and deploy the trained model as a serverless endpoint for sentiment analysis.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and Runpod resources are declared above. Replit supported development and prototype hosting; Runpod supported GPU model training and serverless inference.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype processes complaint-related case data such as customer names, email addresses, messages, order references, ratings, and resolution records where supplied. API credentials are stored as environment secrets and are not placed in source code or this declaration. Training materials used for the sentiment model are review datasets; their provenance and licences require final confirmation. The prototype should use only authorised or demo data during judging. Retention settings and production access controls must be defined before any real customer deployment.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

CoSolve depends on external services: Runpod for sentiment inference, Grafilab for language-model assistance, Clerk for staff authentication, and its Replit deployment configuration. These services may have availability, quota, latency, credential, or configuration failures. The sentiment classifier can be less reliable on neutral, short, or ambiguous complaints, while language-model response drafts can be incomplete or unsuitable. CoSolve displays sentiment confidence, flags uncertain results for human review, requires authenticated staff access, keeps an audit trail, and requires staff approval before consequential actions. It is a decision-support prototype, not an autonomous refund or messaging system.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



23/10/2026